RMI AND CORBA

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INTRODUCTION:

Both RMI (Remote Method Invocation) and CORBA (Common Object Request Broker Architecture) are distributed computing technologies that provide communication to objects on devices on the network.

RMI ARCHITECTURE:

- Client and server communicate via **Java RMI** protocol.
- Client makes a method call to a remote object, which the server implements.

CORBA ARCHITECTURE:

- CORBA uses **IDL (Interface Definition Language)** to define object interfaces.
- Objects communicate across different programming languages and platforms via **IIOP**.

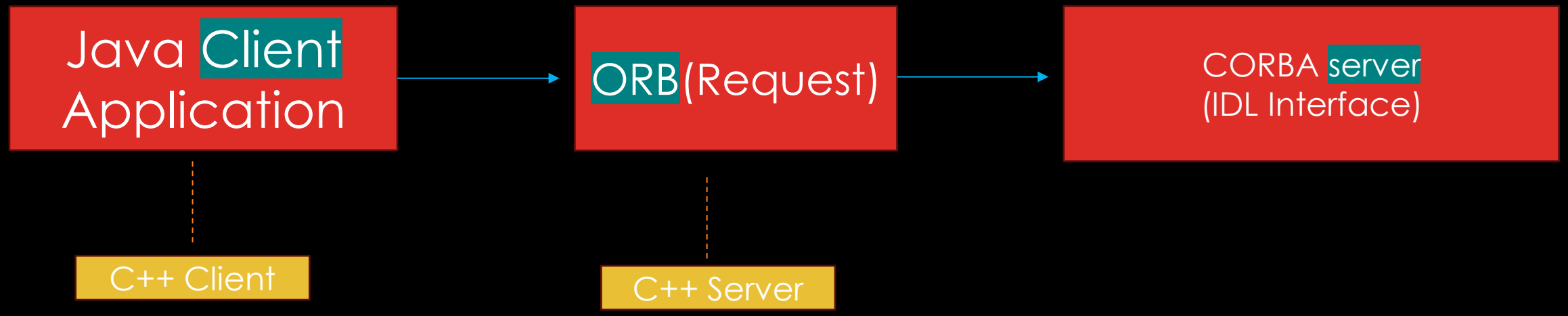
RMI DIAGRAM:

- Client calls method on a **Remote Object** using **Java RMI Protocol**.
- Server has a **Remote Object Implementation**.
- Simple and Java-specific communication mechanism.

CORBA DIAGRAM:

- Clients in different programming languages (e.g., Java, C++) communicate through **IIOP**.

- **ORB** (Object Request Broker) handles the interaction between client and server across platforms.



KEY DIFFERENCES:

Feature	RMI	CORBA
Technology Type	Java-specific technology	Language-independent (supports multiple languages)
Architecture	Simple and easy-to-use for Java	Complex and flexible, supports a wide range of languages
Protocol	Uses Java-specific protocols (JRMP, IIOP)	IIOP (Internet Inter-ORB Protocol)
Object Model	Based on Java objects	Based on CORBA objects (IDL-based)
Performance	Generally faster for Java-to-Java communication	More overhead due to language independence
Security	Basic Java security mechanisms	Rich security features (e.g., SSL, Authentication)
Use Cases	Primarily Java-based applications	Enterprise applications across heterogeneous environments

ADVANTAGE AND DISADVANTAGE

- ADVANTAGE:
- **RMI**: Simpler to implement in a Java-only environment, faster performance within Java ecosystems.
- **CORBA**: Platform and language independent, offers greater flexibility in heterogeneous environments.
- DISADVANTAGE:
- **RMI**: Limited to Java, not suitable for applications that need to support multiple languages or platforms.
- **CORBA**: More complex and may involve higher overhead for smaller-scale applications.

SUMMARY:

- **RMI** is simpler, Java-centric, and uses Java-specific protocols.
- **CORBA** is more complex but supports heterogeneous systems and multiple languages.

THANK YOU !!!!